

Honors Calculus III, Homework 2

Please upload solutions to **Question 1(a, b, c) and Question 2** by **8pm on Friday, April 7**. You can use any results you want from lectures, provided you state them clearly and you are not explicitly asked to prove them in the question, but make sure to write complete proofs. The remaining problems will not be graded, and you should not hand them in, but you are still expected to learn how to do them, unless it says “optional”. As usual, for full credit you need to show and explain your work.

1. (a) (5pt) Let $f_n : X \rightarrow \mathbf{R}$ be a sequence of bounded functions on a set X of real numbers. Assume that $f_n \rightarrow f$ uniformly on X . Prove that f is bounded.
- (b) (5pt) Prove that

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n} \tag{0.1}$$

converges pointwise to $\log(1+x)$ on $(-1, 1)$. (Hint: you could start by estimating the remainder in Taylor’s theorem for $\log(1+x)$, or by differentiating (0.1) termwise.)

- (c) (5pt) Deduce that (0.1) does not converge uniformly on $(-1, 1)$.
 - (d) More generally, assume that a power series converges pointwise to an unbounded function on a bounded interval. Prove that the convergence is not uniform.
2. (5pt) Let $F = \sum_{n \geq 0} \alpha_n x^n$ be a power series. Assume that the sequence

$$\{|\alpha_n/\alpha_{n+1}|\}$$

converges to a limit R (which can be either finite or infinite). Use the ratio test to prove that R is the radius of convergence of F .

(Hint: you are supposed to prove that the series converges if $|x| < R$ and diverges if $|x| > R$.)

3. Give examples (with proof) of power series F, G and H with radius of convergence 1 such that
 - (a) F converges at $x = 1$ and $x = -1$,
 - (b) G converges at $x = 1$ but not at $x = -1$,
 - (c) H does not converge at $x = 1$ and does not converge at $x = -1$.

4. Spivak, *Calculus*, Chapter 24: Problems 13, 14, 25, 26.

Solutions to 1. (1a) Let $\epsilon = 1$. By definition of uniform convergence there exists $n > 0$ such that

$$|f_n(x) - f(x)| < 1 \text{ for all } x \in X.$$

Since f_n is bounded, there exists $M > 0$ such that $|f_n(x)| < M$ for all $x \in X$. By the triangle inequality we conclude that

$$|f(x)| \leq |f(x)| + |f_n(x) - f(x)| \leq M + 1$$

and so f is bounded on X .

(1b) Let

$$F(x) = x - \frac{x^2}{2} + \frac{x^3}{3} + \cdots = \sum_{n \geq 1} (-1)^{n+1} \frac{x^n}{n}.$$

By the ratio test, this series converges absolutely if $|x| < 1$. Hence its radius of convergence is at least one (indeed, it is exactly one since this series diverges at $x = -1$). It follows from this and a theorem from class that the power series $F(x)$ defines a differentiable function $f(x)$ on the interval $(-1, 1)$, and that the derivative of f is given by differentiating $F(x)$ term by term:

$$f'(x) = 1 - x + x^2 - x^3 + \cdots.$$

This is the geometric series for $(1+x)^{-1}$, and so we deduce that

$$f'(x) = \frac{1}{1+x}$$

for all $x \in (-1, 1)$. Hence there exists a constant c such that

$$f(x) = \log(1+x) + c$$

and evaluating at $x = 0$ we find that $c = 0$. So $F(x)$ converges pointwise to $\log(1+x)$ on $(-1, 1)$.

(1c) The partial sums $s_n(x)$ of $F(x)$ are polynomial functions on $(-1, 1)$, so they are continuous on the larger interval $[-1, 1]$: since this is closed and bounded, it follows that s_n is bounded on $[-1, 1]$ for all n . Hence it is also bounded on $(-1, 1)$.

We know from part (b) that $\sigma_n \rightarrow \log(1+x)$ pointwise. If the convergence were uniform, it would follow from part (a) that $\log(1+x)$ is bounded on $(-1, 1)$. Since this is not true, it follows that the convergence is not uniform.

(1d) The same argument of (1c) works, since the partial sums of any power series are polynomials, and so are bounded on every bounded interval.

Solutions to 2. For any real number x we have

$$\lim_{n \rightarrow \infty} \left| \frac{\alpha_{n+1} x^{n+1}}{\alpha_n x^n} \right| = |x| \lim_{n \rightarrow \infty} \left| \frac{\alpha_{n+1}}{\alpha_n} \right| = \frac{|x|}{R}.$$

By the ratio test, it follows that if $|x| < R$ then $F(x)$ converges absolutely.

There remains to prove that if $|x| > R$ then $F(x)$ does not converge. So assume that $F(x)$ converges and choose $\rho \in (R, |x|)$. The ratio test implies that $F(\rho)$ does not converge, since $\rho/R > 1$. However, a theorem from class implies that if $F(x)$ converges then $F(z)$ converges absolutely whenever $|z| < |x|$: see Spivak, Chapter 24, Theorem 6. This is a contradiction since $|\rho| < |x|$.

Solutions to 3. (3a) An example is given by

$$F(x) = \sum_{n \geq 1} \frac{x^n}{n^2}.$$

We have

$$\lim_{n \rightarrow +\infty} \frac{|1/n^2|}{|1/(n+1)^2|} = 1$$

so the radius of convergence of F is one, by Problem 2. However, the series $\sum_{n \geq 1} 1/n^2$ converges, and so $F(x)$ converges absolutely at both $x = 1$ and $x = -1$. The function defined by $F(x)$ is called the *dilogarithm*.

(3b) By Leibniz's theorem and Problem 1, the Taylor series at zero of the function

$$G(x) = \log(1+x)$$

works.

(3c) If $G(x)$ answers part (b), then $H = -G$ answers part (c).